Cybersecurity - Graduate Level Assessment 24 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. Which of the following is not a transport layer vulnerability?
   (a) Mishandling of undefined, poorly defined variables
   (b) The Vulnerability that allows "fingerprinting" & other enumeration of host information
   (c) Overloading of transport-layer mechanisms
   (d) Unauthorized network access

2. Encryption and decryption provide secrecy, or condentiality, but not
   (a) Authentication
   (b) Integrity
   (c) Privacy
   (d) All of the above

3. A/an ________________ is a program that steals your logins & passwords for instant messaging applications.
   (a) IM - Trojans
   (b) Backdoor Trojans
   (c) Trojan-Downloader
   (d) Ransom Trojan

4. Which of the following styles of fuzzer is more likely to explore paths covering every line of code in the following program?
   (a) Generational
   (b) Blackbox
   (c) Whitebox
   (d) Mutation-based

5. The AH Protocol provides source authentication and data integrity, but not
   (a) Integrity
   (b) Privacy
   (c) Nonrepudiation
   (d) Both A & C

True / False (10 marks)

Answer True or False

6. True or False: A session symmetric key can be reused multiple times across different sessions without compromising security.

7. True or False: WPA exclusively uses AES encryption, rendering TKIP obsolete in all scenarios.

8. True or False: Message authentication is a service that ensures message integrity by verifying the authenticity of the message origin.

9. True or False: CBF is not a recognized block cipher operating mode within cryptographic standards.

10. True or False: Buffer-overflow issues can be effectively mitigated by comprehensive memory checks alone without boundary checks.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the role of the Tor browser as a specialized tool for accessing the Tor network, focusing on its significance for anonymous internet users and the implications for cybersecurity.

12. Discuss the role of address sanitizer (ASAN) in enhancing the effectiveness of fuzzing techniques for detecting memory errors in programs. How does ASAN facilitate error identification?

End of Paper